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Robust optimization to secure urban bulk water supply against extreme drought and uncertain climate change☆

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ABSTRACT

Urban bulk water systems supply water with high reliability and, in the event of extreme drought, must avoid catastrophic economic and social collapse. In view of the deep uncertainty about future climate change, it is vital that robust solutions be found that secure urban bulk water systems against extreme drought. To tackle this challenge an approach was developed integrating: 1) a stochastic model of multi-site streamflow conditioned on future climate change scenarios; 2) Monte Carlo simulation of the urban bulk water system incorporated into a robust optimization framework and solved using a multi-objective evolutionary algorithm; and 3) a comprehensive decision space including operating rules, investment in new sources and source substitution and a drought contingency plan with multiple actions with increasingly severe economic and social impact. A case study demonstrated the feasibility of this approach for a complex urban bulk water supply system. The primary objective was to minimize the expected present worth cost arising from infrastructure investment, system operation and the social cost of "normal" and emergency restrictions. By introducing a second objective which minimizes either the difference in present worth cost between the driest and wettest future climate change scenarios or the present worth cost for driest climate scenario, the trade-off between efficiency and robustness was identified. The results show that a significant change in investment and operating strategy can occur when the decision maker expresses a stronger preference for robustness and that this depends on the adopted robustness measure. Moreover, solutions are not only impacted by the degree of uncertainty about future climate change but also by the stress imposed on the system and the range of available options.

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1. Introduction

In the review of the challenges to better solve water resources problems using evolutionary algorithms, Maier et al. (2014) observe that uncertainty affects all aspects of water resources management and that the key sources of uncertainty need to be made "visible" to the optimization process in order to ensure the optimizer does not "have an overinflated sense of control and produce solutions that are likely to be suboptimal and possibly reckless". There are different types of uncertainty, of which deep or severe uncertainty is particularly challenging to handle because it is not amenable to conventional statistical analysis. This study explores the challenge of finding robust optimal solutions that secure real urban bulk water systems against extreme drought in the presence of deep uncertainty about future climate change.

Recent studies on impacts of climate change on water resources have highlighted the potential threat of shifting climate on urban water supply (Brekke et al., 2009; Buytaert and De Bièvre, 2012; Cha et al., 2012; Raje and Mujumdar, 2010; Vicuna et al., 2010). With the prospect of continuing population growth in major cities, the provision of secure water supply will become more pressing and, even more so, if the future climate becomes drier.

Despite advances in climate modelling over the last decade the science of climate change is such that the accuracy of model projections of future climate change is limited (e.g. Randall et al., 2007; Stainforth et al., 2007; Koutsoyiannis et al., 2008, 2009; Blöschl and

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Montanari, 2010; Montanari et al., 2010). Moreover, the absolute likelihoods of future climate change scenarios cannot be determined (Brekke et al., 2009). This uncertainty about future climate change is described as "deep uncertainty" because the planning for secure urban water supply involves "parties to a decision [who] do not know or do not agree on the system model(s) relating actions to consequences or the prior probability distributions for the key input parameters to those model(s)" (Hall et al., 2012, p.1665).

Urban bulk water systems are typically required to supply water with high reliability and, in the event of extreme drought, maintain sufficient supply to enable the urban area to avoid catastrophic economic and social collapse. They typically have a complex infrastructure network that harvests water from multiple surface and groundwater sources, stores water in reservoirs, treats and transfers water to consumption zones. The decision space is complex and nonlinear and often there are multiple conflicting objectives. Mortazavi et al. (2012) showed that coupling a Monte Carlo simulation model based on network flow programming with multi-objective evolutionary optimization offers the prospect of finding practically useful optimal solutions for complex urban bulk water systems. They used historical data to calibrate a stochastic model for generating long time series to ensure the bulk water system was subjected to extreme drought. Of particular relevance here was their finding that the cost of optimal future investment portfolios can be very sensitive to the return period of the design drought (that is, the most severe drought the system can survive without collapse) and that the bulk water system can be extremely vulnerable to droughts with return periods in excess of the design return period.

A significant shortcoming of the Monte Carlo approach used by Mortazavi et al. (2012) was that it assumed the instrumental hydrological record is representative of the future. Given the prospect of anthropogenic-induced climate change impacting water availability, conditioning the optimization of the urban bulk water system on the assumption of a stationary climate runs the risk of exposing the system to unwanted vulnerability in the event that the climate in the planning period tracks a trajectory less favourable than that assumed. This study seeks to address this problem of dealing with "broken assumptions" (Weaver et al., 2013) arising from deep uncertainty about future climate change. Its main contribution is to fuse a number of existing approaches to demonstrate a methodology that helps the decision maker identify the operating rules, drought contingency plan and investment portfolio that trade-off sensitivity to "broken" assumptions about uncertain future climate change against expected economic efficiency. Its key contributions in the context of urban bulk water planning are: 1) the incorporation of the Monte Carlo approach of Mortazavi et al. (2012) into the robust optimization framework of Mulvey et al. (1995) with deep uncertainty about future climate change scenarios; 2) the construction of extreme drought sequences under different future climate change trajectories; and 3) the use of a comprehensive decision space that integrates decisions controlling system operation, determining investment in new sources and source substitution, and defining the drought contingency plan that represents a set of staged decisions responding to a drought emergency.

The paper is organized as follows: Section 2 reviews the concept of robustness and robust optimization and then selects a robust optimization framework that facilitates the exploration of the trade-off between efficiency and sensitivity to different climate change scenarios. Section 3 presents the case study involving an Australian urban bulk water system with a complex decision space involving a comprehensive drought contingency plan involving normal and emergency rationing, source augmentation and substitution, operating rules and capital investments. It describes the construction of a stochastic model to generate long sequences of multi-site streamflow representative of a plausible range of future climate change scenarios. Section 4 presents and discusses results for two demand scenarios using two different measures of robustness and different decision spaces. Section 5 presents the conclusions.

2. Robustness and robust multi-objective optimization

Urban water agencies are tasked with the responsibility of planning and operating bulk water systems in a way that minimizes economic, social and environmental costs while providing an acceptable (and usually very high) level of drought security. The management of urban drought security typically involves a two-pronged strategy:

1. *Risk mitigation*: This involves development of medium to long-term strategies that affect water use efficiency and behaviours, and long-lead time water source infrastructure associated with surface and subsurface water storage, harvesting and recycling, to manage the risk exposure to severe drought.
2. *Drought contingency*: Once a drought develops, trigger events, whose probability of occurrence is determined by the risk mitigation strategy, initiate short-term responses such as restrictions/rationing and short-lead time (and usually very expensive) source augmentation.

The number of feasible solutions can be very large. The typical goal is to identify solutions which maximize drought security, minimize operating and investment costs and minimize social impacts. Because these objectives are conflicting, there is no one best solution. Multi-objective optimization identifies the Pareto optimal set of solutions where each solution cannot be improved with respect to any objective without worsening at least one other objective (Deb, 2001). The Pareto optimal set presents the optimal trade-offs between competing objectives.

However, use of multi-objective optimization in itself is not sufficient. If the optimal solutions are based on assumptions about the future trajectory of the system about which there is deep uncertainty, risk-averse decision makers will shun solutions that are optimal for a particular trajectory but produce poor or unacceptable outcomes for other plausible trajectories – McInerney et al. (2012, p.549) liken such optimality as "dancing on the top of a needle". Considering the risks posed by climate change, Matalas and Fiering (1977) introduced the concept of robustness to the water resources field, describing it as 'the insensitivity of a system design to errors, random or otherwise, in the estimate of those parameters affecting design choice'. Robust solutions should be found so that they can be adaptable to a range of "wait and see" strategies "with some economic efficiency or optimality traded in favour of adaptability and robustness" (Matalas and Fiering, 1977). Dessai and Hulme (2007) argued that decisions will and must continue to be made even "in the absence of accurate and precise climate predictions". In the face of deep uncertainty, they argue that overreliance on projections made by climate models is unwise. Consequently, solutions should be robust across a range of possible futures.

There are several definitions of robustness. However, Hall et al. (2012, p.1657) observe that "most capture the idea of satisficing over many plausible future states of the world". Moreover, there are number of different approaches to making robust decisions. Lempert and Collins (2007) present an insightful case study involving deep uncertainty to compare three robust decision making approaches – the first trades some measure of optimal performance in return for less sensitivity to different plausible futures, the second for satisficing solutions which produce acceptable

performance across the range of plausible futures, while the third seeks to keep options open in a sequential decision process. Of interest was the conclusion that the three approaches were "found to identify similar strategies as the most robust choice" (Lempert and Collins, 2007, p.1009). Hall et al. (2012) compare two widely used approaches for making robust decisions, the info-gap method of Ben-Haim (2006) and the robust decision making framework of Lempert et al. (2003). They observe that "given the diversity of definitions of robustness, and the differing judgements called for in implementing alternative robust decision methods, it is perhaps surprising they often reach similar results" (Hall et al., 2012, p.1671). These comparative studies, though not exhaustive, suggest that the selection of an approach to making robust decisions that is well suited to the problem context and the preferences of the decision maker is an important consideration.

2.1. Selecting a robust multi-objective optimization formulation for urban bulk water systems

Ray et al. (2013) review different methods for optimization under uncertainty. They discuss the differing views on what is robust optimization in the water resource field and conclude with a definition that states that robust optimization is "any optimization technique explicitly incorporating uncertain input data (model parameters) resulting in a trade-off between multiple objectives …" They adopt the robust optimization framework introduced by Mulvey et al. (1995), which implements the idea of satisficing, noting its relevance to water resource systems. In this framework uncertainty about assumptions (or input parameters) is represented by a set of scenarios. The optimization then identifies solutions that defines the trade-off between some measure of expected performance and some measure of sensitivity or robustness over the set of uncertain scenarios.

Watkins and McKinney (1997) presented the first application of the Mulvey et al. framework to water resource problems. By assigning probabilities to different scenarios, they explored two types of robustness, optimality-robustness which identifies solutions that are "close" to optimal for all scenarios, and feasibility-robustness which identifies solutions that remain "almost" feasible for all scenarios. Jia and Culver (2006) developed a robust optimization approach to account for uncertainty about water quality predictions while minimizing pollutant loads reductions. Xu et al. (2013) developed a hybrid interval-robust optimization method and applied it to a real-world case study for planning regional economic and environmental systems under uncertainty in the context of water quality management. Ray et al. (2013) revisited the work of Watkins and McKinney (1997) to describe the importance of asymmetry in risk aversion, a matter of particular relevance to this study.

More recently, Kasprzyk et al. (2013), Kwakkel et al. (2014) and Hamarat et al. (2014) explored the issue of deep uncertainty in a multi-objective context using multi-objective evolutionary algorithms. Kasprzyk et al. (2013) identified approximate Pareto optimal solutions contingent on best estimates of inputs and then assessed their robustness *a posteriori* using robust decision making analysis. On the other hand, Kwakkel et al. (2014) and Hamarat et al. (2014) used Mulvey-type robust optimization where deep uncertainty is described through a set of scenarios to directly identify robust solutions. While robustness was directly embedded in the optimization, its expression was limited to a single metric involving expected performance and some measure of performance variability - Kwakkel et al. (2014) minimized the median value multiplied by its interquartile distance plus one for each criterion, whileHamarat et al (2014) maximized the signal-to-noise ratio for each criterion.

Following Ray et al. (2013), we adopt the robust optimization framework of Mulvey et al. (1995). We define our robust optimization framework as follows: Suppose there are N future scenarios $\{S_j, j = 1, .., N\}$ with probability of occurrence $\{p_j, j = 1, .., N\}$. Each scenario defines a plausible future state of the system and is expressed as particular values of model parameters or forcing time series that affect the performance of the urban bulk water system. In our study, each scenario will describe multiple time series of hydro-climate data corresponding to a future climate change state.

The simulation model of the urban bulk water resource system is expressed as

$$Z_j = M[x, Q(S_j)] \tag{1}$$

where x is the decision vector describing operating and infrastructure variables that can be changed by the optimization algorithm, $Q(S_j)$ is the system input consisting of streamflow, rainfall, water demand and model parameters conditioned on the scenario S_j, and Z_j is the system output.

The robust multi-objective optimization problem therefore becomes:

$$\begin{aligned} &\min_x \left\{ \sum_{j=1}^{N} p_j f_i(x|Z_j),\ R[f_i(x|Z_1), \ldots, f_i(x|Z_N)],\ \ i = 1, .., M \right\} \\ &\text{subject to } Z_j = M[x, Q(S_j)],\ \ j = 1, .., N \\ &g(x) \le 0 \end{aligned} \tag{2}$$

where $f_i(x|Z_j)$ is the value of the ith performance objective for a given decision x and output Z_j conditioned on the jth scenario, and $R[f_i(x|Z_1), \ldots, f_i(x|Z_N)]$ is a robustness measure expressed as a function of the objective values evaluated for the N scenarios. The term $g(x)$ is some function of the decision variables used to constrain the decisions – for example, one may impose the constraint that the urban bulk water system does not experience catastrophic failure (that is, run out of water) in any scenario.

Eqn. (2) describes a one-stage stochastic programming model in the sense that the decision vector x is not conditioned on any particular scenario. In view of the considerable natural variability in climate, it is unlikely that the actual climate change scenario would be identified over the time scale of the analysis.

2.2. Deep uncertainty and subjective probability

In the presence of deep uncertainty about future scenarios, the Mulvey et al. (1995) framework appears, at first sight, to be subject to a contradiction. The framework assumes a known probability can be assigned to each scenario – that is, there is certainty about the scenario probability. However, in the case of future climate change, it is accepted that probabilities cannot be assigned in an objective manner to different scenarios. There is little choice but to assign subjective probabilities.

One way to avoid assignment of subjective probabilities is to adopt the precautionary principle (O'Riordan and Jaeger, 1996) and assume the worst-case scenario. However, this is problematic when dealing with future climate change as it remains unclear what the probability of the worst-case scenario is and even what the worst-case scenario will look like (e.g. Koutsoyiannis et al., 2008, 2009; Blöschl and Montanari, 2010; Montanari et al., 2010; Kiem and Verdon-Kidd, 2011). Lempert and Collins (2007) argue that the precautionary principle, while having intrinsic appeal, offers no way to explore trade-offs and no systematic way to explore the uncertainty about the uncertainty.

The Mulvey et al. (1995) robust optimization framework was developed to give expression to risk-averse behaviour. It remains

relevant even in the case of deep uncertainty because it allows the decision maker to reflect about his/her degree of confidence in the assigned probabilities. The trade-off between some measure of (expected) efficiency and robustness allows reflection about the degree of uncertainty about the uncertainty. If the decision maker has little confidence in the assigned probabilities, he/she is likely to favour solutions that sacrifice efficiency in return for less sensitivity to scenario outcome. Conversely, if the decision maker has reasonably high confidence in the assigned probabilities, he/she is likely to favour solutions with higher efficiency in return for less sensitivity to scenario outcome. McInerney et al. (2012) present a similar justification in a robust optimization case study involving greenhouse gas mitigation strategies in which they compare expected utility maximization with limited-degree-of-confidence and safety-first decision criteria.

2.3. Choice of robustness measure

The robustness measure $R[f_i(x|Z_1),\ldots,f_i(x|Z_N]$ expresses some measure of variability in performance across the scenarios. For example, it may be the standard deviation of the criterion values over the N scenarios (Ray et al., 2013), which depends on the scenario probabilities which, in the case of deep uncertainty, are themselves uncertain. If a decision maker is uncomfortable with the dependence of the robustness measure on scenario probabilities, a conservative approach could be adopted in which the robustness measure is based on worst case outcomes. For example, two conservative measures, independent of scenario probabilities, are the spread defined as

$$R\left[f_i(x|Z_1),\ldots,f_i(x|Z_N\right] = \left|\max_j f_i(x|Z_j) - \min_j f_i(x|Z_j)\right| \quad (3)$$

and the maximum criterion value defined as

$$R\left[f_i(x|Z_1),\ldots,f_i(x|Z_N\right] = \max_j f_i(x|Z_j) \quad (4)$$

While equations (3) and (4) are mathematically independent of scenario probabilities, it would be incorrect to suggest that the scenario probability has been removed from the decision making. Indeed it is very likely the decision maker will reflect on the uncertain probability of sampling the worst case scenario when examining the trade-off between efficiency and robustness. The worst-case robustness measure provides an end point in the sense that it is unlikely that outcomes will get worse. It invites the decision maker to reflect and bet on the possibility that the outcomes will not be as bad as suggested by the worst-case measures. Of course this perspective depends on our certainty that the identified worst-case scenarios indeed represent the worst cases, a topic that will be visited in the discussion of the case study.

The robustness measures given by equations (3) and (4) are intrinsically different. As Ray et al. (2013) observe measures such as the standard deviation and spread are symmetric in the sense they "equally" penalize deviations above or below the expected value. In contrast, the maximum criterion given by equation (4) only penalizes "undesirable" deviations. As this asymmetry could lead to quite different decisions, it is important that the decision maker carefully reflects on the desirable characteristics of a robust solution. In the case study we will consider both measures as they produce very different conclusions.

The efficiency measure $\sum_{j=1}^{N} p_j f_i(x|Z_j)$ explicitly depends on scenario probabilities. In the case of deep uncertainty, this measure should not be seen as an absolute measure of efficiency but rather a reference point for efficiency. In our case study, we assigned equal scenario probabilities to the future climate change scenarios.

2.4. Selection of multi-objective evolutionary algorithm

Earlier applications of the Mulvey et al. (1995) robust optimization framework such as that of Watkins and McKinney (1997) used classical mathematical programming. This imposes a practical limitation on the complexity of the system and the specification of objectives, constraints and decisions. We relax this limitation by solving eqn (2) using a multi-objective evolutionary algorithm.

Evolutionary algorithms (EAs) represent a class of stochastic optimization methods based on the process of natural evolution. They have been employed in a variety of engineering applications and these algorithms have proven themselves as general, robust and powerful methods (Coello Coello et al., 2007; Deb, 2001). EAs reduce many of the restrictions imposed by classical mathematical programming offering a richer set of applications. In water resource optimization studies, EAs have seen wide use – see review by Nicklow et al. (2010).

There are many stochastic optimization algorithms to choose from. In our case study we adopted the ε-multi-objective optimization evolutionary algorithm (εMOEA) described in Mortazavi et al. (2012). This choice was guided by the findings of Mortazavi-Naeini et al. (2014) who compared the performance of εMOEA against several other algorithms including particle swarm and ant colony using two urban bulk water systems as case studies. They found that no algorithm was superior and that εMOEA performed credibly. εMOEA is a member of the evolutionary algorithm family whose distinguishing feature is the use of the ε-dominance concept which divides the objective space into hyperboxes of size ε and allows only one non-dominated solution to reside in each box (Laumanns et al., 2002). Inclusion of this concept in a genetic algorithm (GA) framework produces a method capable of maintaining a diverse and well-distributed set of solutions with a small algorithmic computational cost (Deb et al., 2003). The εMOEA code was developed by authors and can be implemented in a multiple-host, multiple-core environment. Other ε-dominance multi-objective evolutionary algorithms used in water resource applications such as ε-NSGA and BORG are discussed by Reed et al. (2013).

3. Case study: problem formulation

We illustrate the Mulvey et al. (1995) robust optimization approach using a case study based on the Lower Hunter urban bulk water system. This system provides drinking water to over 200,000 properties in the Lower Hunter of New South Wales, Australia. Water is harvested from a combination of surface storages and groundwater resources shown in Fig. 1. Major components of the bulk water system include Chichester Dam, Seaham Weir on the Williams River from which water is pumped via Balickera channel to Grahamstown Dam, an off-river storage, and the Tomago and Tomaree groundwater sources.

The system illustrated in Fig. 1 was setup as the WATHNET5 model (Kuczera et al., 2009) shown in Fig. 2. The groundwater sources were modelled as two lumped storages with head-dependant Darcian-like fluxes between the storages and the ocean. The model included a range of future options including a new reservoir on the Chichester River, desalination and domestic rainwater tanks together with an extensive drought contingency plan. A full discussion of the system and model can be found in Mortazavi-Naeini et al. (2013a,b).

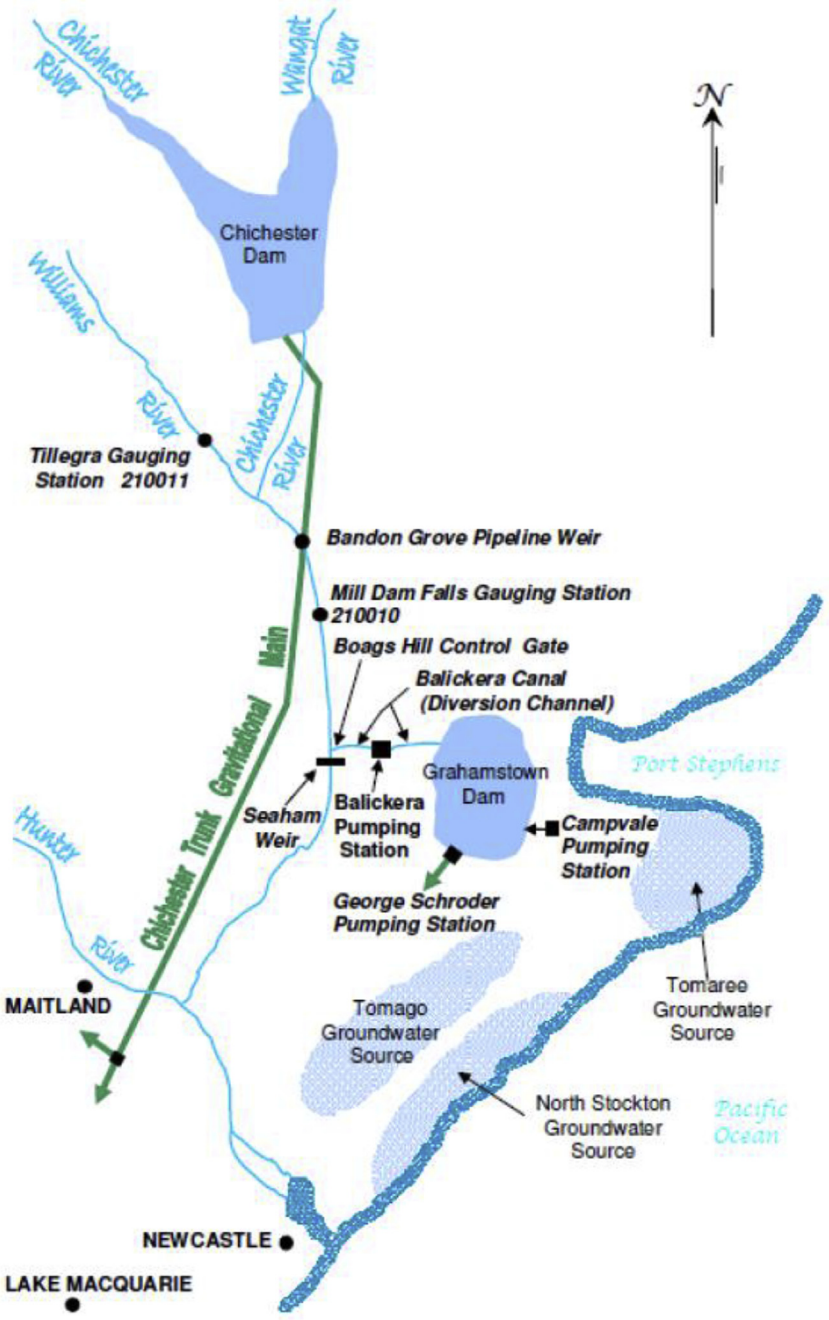


Fig. 1. Map of existing Lower Hunter bulk water system.

3.1. Construction of uncertain future climate change scenarios

The construction of future climate change scenarios has to incorporate the very considerable influence of natural climate variability, the identification problems associated with short hydro-climatological instrumental records, and the concern about how human-induced climate change may increase the frequency and severity of future extreme events, including droughts and floods.

There have been attempts [see Kiem and Verdon-Kidd (2011) for a review] to utilize climate model outputs to determine how anthropogenic climate change may affect water resources and, on the basis of this information, to develop water resource management strategies to deal with the projected risks. However, the uncertainty associated with future climate change projections is known to be significant (e.g. Kiem and Verdon-Kidd, 2011) and is magnified further when attempting to make inferences at the regional scale (e.g., differentiating between coastal and inland processes). This is especially the case for precipitation [e.g. Lim and Roderick (2009) and hydro-climatic extremes (see IPCC, 2012)]. The uncertainty is so high that projections of future drought risk, on either the short (seasonal up to 5 years) or long (more than 10 years into the future) term, currently have limited practical usefulness for water resource managers and/or government policy makers (National Climate Change Adaptation Research Facility, 2010).

However, methods do exist to quantify risks associated with climate variability and/or change that do not rely solely on climate model outputs (e.g., McMahon et al., 2008; Verdon-Kidd and Kiem, 2010). These methods involve utilizing stochastic modelling

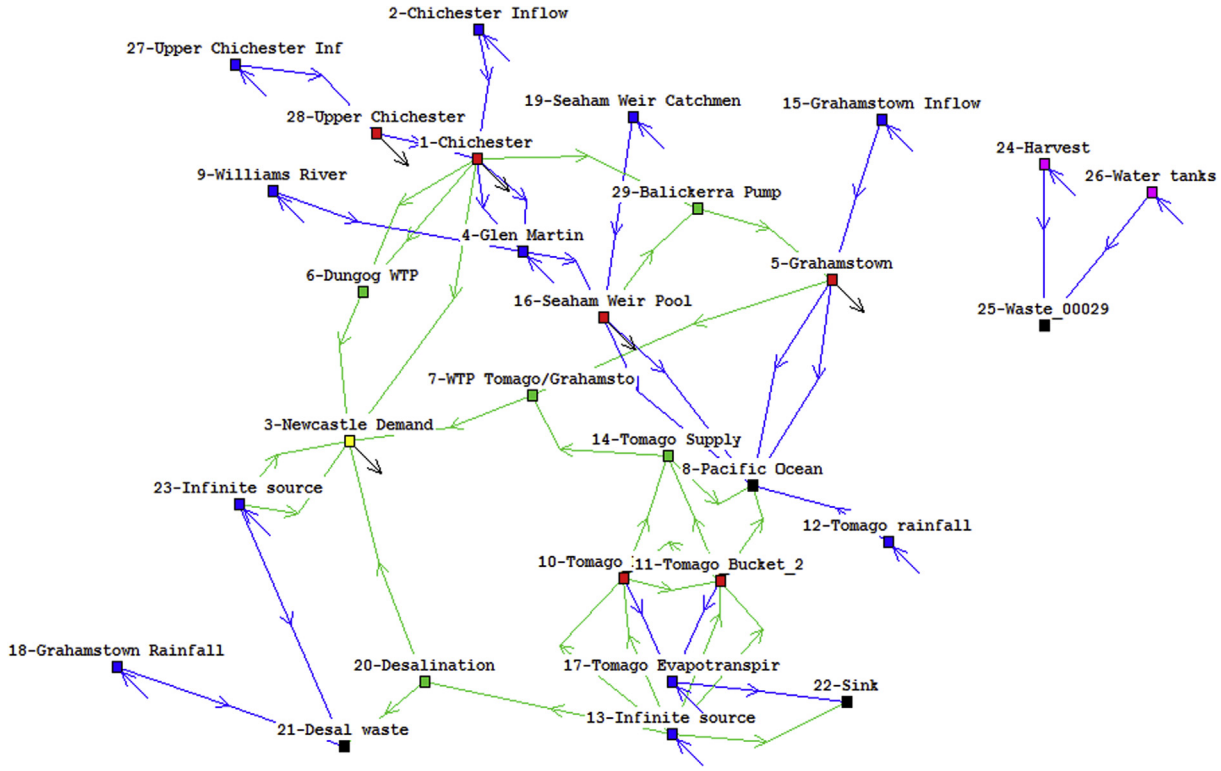


Fig. 2. WATHNET5 schematic of Lower Hunter bulk water system with inclusion of future options. Red nodes represent reservoirs, blue represent inflows, yellow represent demands and green represent junctions. Blue arcs represent streams, while green arcs represent conduits. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

approaches that replicate the important statistics associated with historical data (e.g. interannual to multidecadal natural climate cycles, dry spells, wet epochs, extreme high and low rainfall etc.) and then utilising the best available climate modelling information to provide a range of plausible future scenarios.

This study focuses on the future climate around 2070 (i.e. the average climate during 2060–2079). To simulate the performance of the Lower Hunter system, streamflow replicates representative of the 2070 climate were developed as follows:

1. Select two 2070 climate change scenarios representing the range in likely 2070 climate conditions – these scenarios S_1 and S_2 will be referred to as 'dry' 2070 and 'wet' 2070 respectively.
2. For each scenario, perturb observed historic rainfall and potential evapotranspiration (PET) time series using the selected 2070 climate change factors.
3. Calibrate a stochastic multi-site model of rainfall and PET to the perturbed historic data.
4. Generate 10,000 50-year replicates of daily rainfall and PET representative of the "perturbed" climate.
5. Using calibrated rainfall-runoff models produce 10,000 50-year replicates of monthly streamflow at multiple sites.

The climate change factors used in step (2) were obtained from CSIRO-BoM (2007), which is based on the Coupled Model Intercomparison Project 3 (CMIP3) model results as used in IPCC (2007), and are reported as a range in Table 1. This range was derived as follows:

1. For the A1FI emission scenario, the projected range for global warming based on all CMIP3 general circulation models (GCMs), run several times, is 1.74–4.64 C° in 2070.
2. Using the 50th percentile estimate of local change in rainfall and PET per degree of global warming, a range of climate change factors for the Lower Hunter was obtained. The upper bound for rainfall and lower bound for PET of this range was used to develop the 'wet' 2070 scenario and the lower bound for rainfall and upper bound for PET used to produce the 'dry' 2070 scenario. The 50th percentile was used in recognition of the numerous studies (e.g. Kiem and Verdon-Kidd (2011)) and other more recent work produced as part of the South East Australian Climate Initiative (SEACI, http://www.seaci.org/) that have demonstrated that of the 23 GCMs used to develop future climate projections by Climate Change in Australia (CSIRO-BoM, 2007), fewer than 8 can be considered close to realistic for Australian hydro-climatology and that best practice at the time this work was conducted was to use the projections where there is most agreement across the models rather than the extremes.

3.2. Decisions

A large number of options were available to control the performance of the bulk water system. In this case study, fourteen

Table 1
Climate change factors (% change to 1990 baseline mean) (obtained from CSIRO-BoM (2007), available at http://climatechangeinaustralia.com.au/) using the 2070 'best estimate' (50th percentile) for the A1FI emission scenario.

Variable	Scenario	Winter	Spring	Summer	Autumn
Rainfall	2070 wet	−10	−10	+2	−2
	2070 dry	−20	−20	−2	−5
PET	2070 wet	+12	+4	+8	+12
	2070 dry	+16	+8	+12	+16

decision variables were identified as being potentially important. They are listed in Table 2 where they are classified as infrastructure (which corresponds to a physical asset), operational (which affects the way the system is operated) or drought contingency (which determines the staged response to severe drought to avoid catastrophic "running out" of water).

Fig. 3 provides more detail on the structure of drought contingency plan. The plan consists of a sequence of storage triggers that initiate specific drought responses. As total storage drops the first trigger (yellow zone (in the web version)) the plan imposes "normal" restrictions with the principal focus on restricting outdoor water use. As storage continues to decline, construction of a desalination plant (orange zone (in the web version)) and imposition of emergency rationing (red zone (in the web version)) are triggered. Emergency restrictions would involve drastic measures including pressure reduction and supply cut-off.

Decisions x_1 to x_7 are used to parameterize the drought contingency plan. Decisions x_1 and x_2 determine the five levels of restrictions. Decision x_3 indicates when emergency rationing would be imposed while x_4 defines how much demand should be decreased when emergency rationing is imposed. Decision x_5 determines when the desalination plant should be implemented and decision x_6 defines the desalination plant capacity. Decision x_7 defines the construction time of the desalination plant. A relationship between construction time and pre-build investment for the desalination plant was used – the greater the pre-build investment plan (involving tasks such as feasibility study, land purchase, pilot testing, approvals and so on), the shorter the construction lead time.

Three infrastructure decisions are considered. Decision x_9 determines the capacity of a new reservoir on the Chichester River, x_{10} determines the expansion in the Chichester to Newcastle pipeline capacity and decision x_{14} defines the number of domestic rainwater tanks installations.

There are four operational decisions. Decision x_8 defines the desalination plant trigger while x_{13} defines the Grahamstown pumping trigger. Decisions x_{11} and x_{12} define the priority for storing water in Grahamstown. All the reservoirs in the WATHNET5 network linear program were assigned 20 carryover arcs which "store" water for the next time step. Each carryover arc has a capacity equal to 1/20 of the reservoir capacity and a gain (i.e. negative cost) defined by

$$Gain(j) = BG + (j-1)*IG, \quad j = 1, \ldots, 20 \tag{5}$$

where BG is the base gain and IG is the incremental gain. On account of Grahamstown's dominant storage, all reservoirs except Grahamstown were assigned a base gain of 10,000 and an incremental gain of 100 – this ensures that reservoirs rise and fall in unison. Decisions x_{11} and x_{12} define the base and incremental gain for Grahamstown respectively. Depending on the values assigned to x_{11} and x_{12}, water may be preferentially stored in Grahamstown or in the rest of the system.

The case study does not consider the full suite of possible decisions available to manage the Lower Hunter bulk water system as the object of the study is to demonstrate the robust optimization framework.

3.3. Objectives and constraints

Current urban bulk water planning practice in New South Wales emphasizes the need to find minimum cost solutions subject to satisfying drought security constraints. To maintain consistency with this practice, present worth cost was selected as the performance (or efficiency) measure. Two robustness measures were considered giving the following objectives:

1. Minimize the total expected present worth (EPW) cost consisting of capital and operating costs and costs of normal restrictions and emergency rationing; and

2a. Minimize the present worth cost spread defined as the difference between the total EPW cost for the dry (S_1) and wet (S_2) climate change scenarios; or

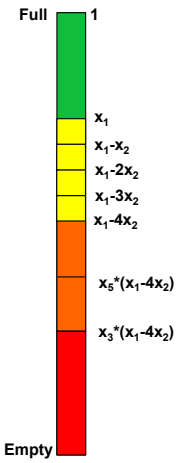
2b Minimize the maximum present worth cost experienced in the dry (S_1) climate change scenario.

The capital cost represents the cost of building new infrastructure, which in this case study, may be a new reservoir, capacity expansion of the pipeline from Chichester to Newcastle and a new desalination plant. Table 3 summarizes the capital costs for the new reservoir and desalination plant. These costs are "indicative" and should not be used in other contexts. The capital cost model uses a binary function: if the asset is selected by the optimization, then the total cost is the sum of a fixed setup cost and cost proportional to the size of the asset; however, if the asset is not selected, the capital cost is zero. For large desalination plants with installed capacity in excess of 20 ML/day, the construction time or lead time varies between 12 and 60 months depending on pre-build investment with shorter lead times requiring greater pre-build investment; if a zero-lead time is selected, the desalination plant is assumed commissioned at the start of the simulation. In contrast, the small desalination plant costs are based on packaged plant designs and have short lead times.

Operating costs, summarized in Table 4, include the cost of pumping transfers from Seaham to Grahamstown reservoir and

Table 2
List of decision variables.

Decision X	Description	Lower limit	Upper limit	Category
1	Level 1 restriction storage trigger	0.00	0.90	Drought contingency
2	Restriction trigger increment	0.05	0.25	
3	Emergency restriction trigger	0.0	1.0	
4	Emergency rationing demand scaling factor	0.4	0.7	
5	Storage trigger to initiate desalination plant construction	0.0	1.0	
6	Desalination plant capacity (ML/day)	0	500	
7	Desalination plant construction lead time (month)	0, 12, 24, 36 or 60		
8	Desalination plant trigger	0.0	1.0	Operational
9	New reservoir capacity (ML)	Either 0 or in range 100,000 to 230,000		Infrastructure
10	Chichester to Newcastle pipeline capacity expansion (ML/day)	0	50	Infrastructure
11	Grahamstown base gain	8000	12,000	Operational
12	Grahamstown incremental gain	20	200	Operational
13	Grahamstown pumping trigger	0.0	1.0	Operational
14	Number of domestic rainwater tanks	0	100,000	Infrastructure



Total storage trigger fraction	Action	Consequence
x_1	Impose level 1 restrictions	Prepare community
x_1-x_2	Impose level 2 restrictions	3% demand reduction
x_1-2x_2	Impose level 3 restrictions	10% demand reduction
x_1-3x_2	Impose level 4 restrictions	20% demand reduction (zero outdoor water use)
x_1-4x_2	Impose level 5 restrictions	28% demand reduction
x_5*(x_1-4x_2)	Commence construction of desalination plant with capacity x_6 [ML/day]	Time to desalination plant operation x_7 months which depends on pre-build investment
x_3*(x_1-4x_2)	Impose emergency restrictions	Total water use reduced by 100*x_4 % where x_4 lies between 0.4 and 0.7 Expect hardship, supply cuts, pressure reductions

Fig. 3. Drought contingency plan.

Table 3
Capital cost summary.

Decision variable	Fixed and unit cost	
New water source reservoir	\$150m + \$1300/ML storage in excess of 100,000 ML	
Small desalination plant (<20 ML/day installed capacity)	\$6m + \$3 m/ML/day installed capacity Pre-build cost = \$0	
Large desalination plant (>20 ML/day installed capacity)	\$150m + \$7.5 m/ML/day installed capacity	
	Lead time months	Pre-build cost as % of total cost
	0	70%
	12	50%
	24	35%
	36	10%
	60	0
Chichester to Newcastle pipeline	\$1.45 m/ML/day installed capacity	
Rainwater tanks	\$5000 per tank	

Grahamstown reservoir to Newcastle. They also include the cost of treating water in water treatment plants and operation of the desalination plant.

Table 5 summarizes the estimated costs of implementing restrictions by the water agency and the social costs borne by the community. The social cost represents the loss of revenue to the water agency and loss of consumer surplus.

Emergency rationing is triggered after the imposition of level 5 restrictions which enforce a total ban on all outdoor water use and encourage other voluntary savings. Because emergency rationing would impose much severer cutbacks to water use, it is expected social costs would climb steeply. Table 6 summarizes indicative social costs which are linearly scaled between the 40 and 70% demand reduction limits. A more considered approach to estimating such costs would involve carefully designed surveys of willingness-to-pay – this was beyond the scope of this study.

Table 4
Summary of operating costs.

Operations	Cost
Pumping from Seaham to Grahamstown reservoir	\$12/ML
Pumping from Grahamstown reservoir to Newcastle	\$76/ML
Tomago water treatment plant	\$80/ML
All other water treatment plants	\$50/ML
Desalination plant	\$1200/ML

Two constraints were imposed on the solutions:

1) The system was not permitted to "run out of water" in any of 10,000 50-year replicates of dry and wet climate. While this may seem to be a very conservative requirement it needs to be stressed that the drought contingency plan allows emergency rationing which would be judged draconian from a social and political perspective. The intent of this constraint is to ensure there is an extremely low risk (circa 1 in 500,000 years) of a catastrophic failure to supply which could lead to economic and social collapse.
2) In any 50-year replicate the time spent in emergency rationing must be no greater than 10%. This constraint effectively puts a limit of a maximum of 5 years of emergency rationing within any 50-year replicate. It guides the optimisation away from solutions imposing a permanent state of emergency rationing.

3.4. Evolutionary algorithm settings

εMOEA is a probabilistic optimization method. Therefore, it is unable to guarantee convergence to the Pareto front. Accordingly, to reduce the chance of premature convergence affecting the results, each scenario was optimized four times with different random number seeds – each run took about 16 h on a 72-core

Table 5
Restriction cost summary.

Restriction level	Financial cost to water agency		Social cost (\$/person/month)
	Fixed (\$)	Variable (\$/month)	
Level 1	297,500	154,000	0
Level 2	70,500	229,250	3.00
Level 3	0	293,250	6.10
Level 4	18,000	305,467	8.02
Level 5	18,000	317,083	10.80

Table 6
Costs associated with emergency restrictions.

Reduction in total demand	Financial cost to water agency		Social cost (\$/person/month)
	Fixed (\$)	Variable (\$/month)	
40%	20,000	350,000	50
70%	100,000	500,000	150

Linux network. The results presented in the next section are the non-dominated solutions obtained from the pooled set of four runs. Based on the optimization studies in Mortazavi et al. (2012, 2014), we used set εMOEA parameters for probability of crossover = 1 and probability of mutation = 0.01. The maximum number of iterations was set equal to 20,000. The εMOEA epsilon values were set to 100,000 to be sufficiently small to ensure high resolution. We note that in practice larger epsilon values would be used – apart from decreasing computational time, coarser resolution of the Pareto front reduces the chance of overwhelming decision makers without any practical loss of information (Kollat and Reed, 2007).

4. Case study: results and discussion

4.1. Demand scenario selection

While the focus of this study is on the deep uncertainty associated with future climate change affecting the 2070 climate, there are many other uncertainties that could be considered when exploring robust solutions. The most significant of these is likely to be uncertainty about future demand. Paton et al. (2013) show in a case study for the southern Adelaide bulk water system that demand was the greatest source of future uncertainty, though its relative contribution to total uncertainty diminished through time.

The methodology described in section 2 is not restricted to climate change scenarios. Each scenario could be constructed to represent a future climate change state and a future level of demand. However, given the focus of this study is on deep uncertainty about future climate change, we chose not to take this path. Our study focusses on the urban system for an uncertain 2070 climate, but does not define the pathway from the present to 2070. Mortazavi-Naeini et al. (2013a) demonstrate in principle how such a pathway can be constructed using a robust multi-stage optimization formulation for an urban bulk water system experiencing uncertain demand growth.

Results are presented for two demand scenarios. The first considers the expected demand in 2060, the most distant forecast currently available. This demand level represents 128% of current demand and is due to a low expected growth annual rate of about 0.5%. The second demand scenario is hypothetical and considers a doubling of current demand. The purpose of this scenario is to provide insight about how our perceptions of robust decisions change as the system moves to a highly stressed state.

For each demand scenario, it is important to stress that the optimization is seeking to throw light on the questions, what are the trade-offs between efficiency and robustness and what are the characteristics of robust decisions for an uncertain 2070 climate. It also needs to be stressed that the simulation is statistically evaluating performance for a climate representative of conditions that could be encountered in 2070.

4.2. Using spread as a robustness measure for 128% × current demand scenario

This section presents results for the 128% × current demand scenario where the expected present worth (EPW) cost and the spread in EPW costs between the dry and wet climate change scenario are minimized. Fig. 4 presents two sets of approximate Pareto-optimal solutions. The first set is produced when all the decisions in Table 2 including the reservoir option are active. In this case, Fig. 4 shows that there is a limited trade-off between total EPW cost and robustness. The total EPW cost ranges from $360m to $485m, while the cost spread across the dry and wet 2070 climate scenarios ranges from $0.08m to $13.4m. The proposition of increasing robustness by reducing the spread from $13.4m to $0.08m for an increased community investment of $125m is not compelling. A closer inspection of the results in Table 7 for the labelled Pareto-optimal solutions reveals that the chance of restrictions is always less than 1.7% regardless of climate change scenario and the chance of emergency restrictions is at worst 0.08%. There is little difference between the solutions other than an increase in the new reservoir capacity; all solutions excluded rainwater tanks and desalination from their portfolios. Overall, there is little sensitivity to future climate change uncertainty.

However, the situation changes radically if the new reservoir option is removed from the decision space – this corresponds to setting decision 9 to zero. The total EPW cost increases significantly and ranges from $812m to $3460m, while the cost spread across

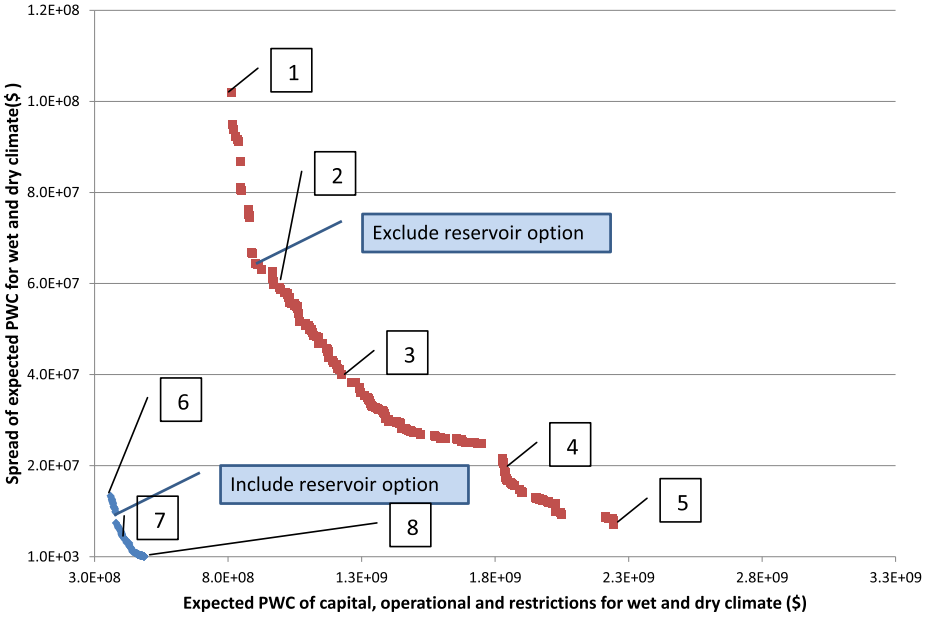


Fig. 4. Robust Pareto optimal fronts for the 128% × current demand scenario.

Table 7
Summary of labelled solutions on Pareto front in Fig. 4 for case when the reservoir option is *excluded*: 128% × current demand scenario.

	Solution	1	2	3	4	5
Objectives	Total expected PW of capital, operational and restriction costs for wet and dry climate ($m)	812	972	1225	1833	3399
	Spread of expected PW cost between dry and wet climate ($m)	102	61	40	20	7
Decisions	Level 1 restriction storage trigger (% of total storage)	27.6	23.6	22.5	9.9	0.7
	Restriction trigger increment (% of total storage)	5.4	5.5	5.3	20.4	15.4
	Emergency restriction trigger	0.774	0.852	0.853	0.906	0.857
	Severe rationing demand scaling factor	0.698	0.661	0.642	0.642	0.679
	Storage trigger to initiate desalination plant construction	0.427	0.382	0.350	0.147	0.050
	Desalination plant capacity (ML/day)	51	75	82	196	350
	Desalination plant construction lead time (month)	0.000	0.000	0.000	0.000	0.000
	Desalination plant trigger (% of total storage)	42.7	38.2	35.0	14.7	5.0
	Chichester to Newcastle pipeline capacity expansion (ML/day)	93	92	92	93	111
	Grahamstown base gain	9842	9837	9837	9853	10,087
	Grahamstown incremental gain	108	108	110	107	199
	Grahamstown pumping trigger (% of total storage)	99.8	99.8	99.8	99.5	100
	Number of domestic rainwater tanks	264	1678	48,589	259	87,569

the dry and wet 2070 climate change scenarios ranges from $6m to $102m.

To gain insight why the Pareto fronts differ, the solutions labelled in Fig. 4 are compared in Tables 7 and 8 for the case without and with the new reservoir option. When the reservoir option is excluded, the only options that provide a new source of water are desalination and domestic rainwater tanks. Desalination is adopted in all solutions (1–5). The least robust solution opts for a 50 ML/day desalination plant operating whenever storage drops below 43%, while the most robust solution opts for an almost maximum rollout of rainwater tanks together with a much larger 350 ML/day desalination plant which is operated whenever storage drops below 5%. On the other hand, when the reservoir option is included, none of the solutions (6–8) opt for a desalination plant and the number of installed rainwater tanks is negligible; however, the reservoir capacity increases as the solutions become more robust.

4.3. Using spread as a robustness measure for 200% × current demand scenario

This section presents results for the 200% × current demand scenario where expected present worth (EPW) cost and the spread in EPW costs between the dry and wet climate change scenario are minimized. Fig. 5 presents the approximate Pareto-optimal solutions with all the decisions in Table 2 activated. Table 9 presents a summary of the six solutions labelled on Fig. 5. Unlike the 128% × current demand scenario, there is a strong trade-off between efficiency and spread. The total EPW cost ranges from $1092m to $2943m, while the spread ranges from $1m to $455m. For solution 1 the spread represents 42% of the total EPW cost. However, reducing the spread requires a disproportionately large increase in EPW. The reduction of spread by $53m when moving from solution 1 to 2 is offset by an increase in total EPW cost of $73m. Moreover, moving from solution 4 to 5 reduces the spread a further $174m in return for an increase in total EPW cost of $596m, while moving from solution 5 to 6 reduces the spread a further $30m in return for an increase in total EPW cost of a staggering $2975m.

Unlike the Pareto solutions for the 128% × current demand scenario which do not favour the desalination option, the Pareto solution set for the 200% × current demand scenario opts for desalination in all cases. The following features of the Pareto solution set have significant implications for managing the system:

a) All solutions opt for the maximum capacity of the new reservoir. This suggests that if it were technically feasible to increase the capacity of the reservoir, the optimizer would exploit the additional capacity.
b) All solutions adopt desalination with zero lead time meaning the plant is commissioned at the start of the simulation and will supply water whenever total storage drops below the trigger. The pressing need for an additional source of water forces the optimizer to embrace desalination as permanent infrastructure rather than a drought contingency measure. Solutions 1–3 (more efficient, less robust) opt for the small and cheap desalination option, while solutions 4–6 (less

Table 8
Summary of labelled solutions on Pareto front in Fig. 4 for case when the reservoir option is *included*: 128% × current demand scenario.

	Solution	6	7	8
Objectives	Total expected PW of capital, operational and restriction costs for wet and dry climate ($m)	360	392	487
	Spread of expected PW cost between dry and wet climate ($m)	13	6	0.0001
Decisions	Level 1 restriction storage trigger (% of total storage)	28.1	25.3	22.5
	Restriction trigger increment (% of total storage)	5.0	5.0	5.2
	Emergency restriction trigger	0.964	0.931	0.887
	Severe rationing demand scaling factor	0.697	0.673	0.545
	Storage trigger to initiate desalination plant construction	–	–	–
	Desalination plant capacity (ML/day)	0.000	0.000	0.000
	Desalination plant construction lead time (month)	–	–	–
	Desalination plant trigger (% of total storage)	–	–	–
	New reservoir capacity (ML)	100,634	129,457	208,818
	Chichester to Newcastle pipeline capacity expansion (ML/day)	85	85	85
	Grahamstown base gain	8995	9748	9853
	Grahamstown incremental gain	32	21	26
	Grahamstown pumping trigger (% of total storage)	98.7	98.8	100
	Number of domestic rainwater tanks	0	0	12

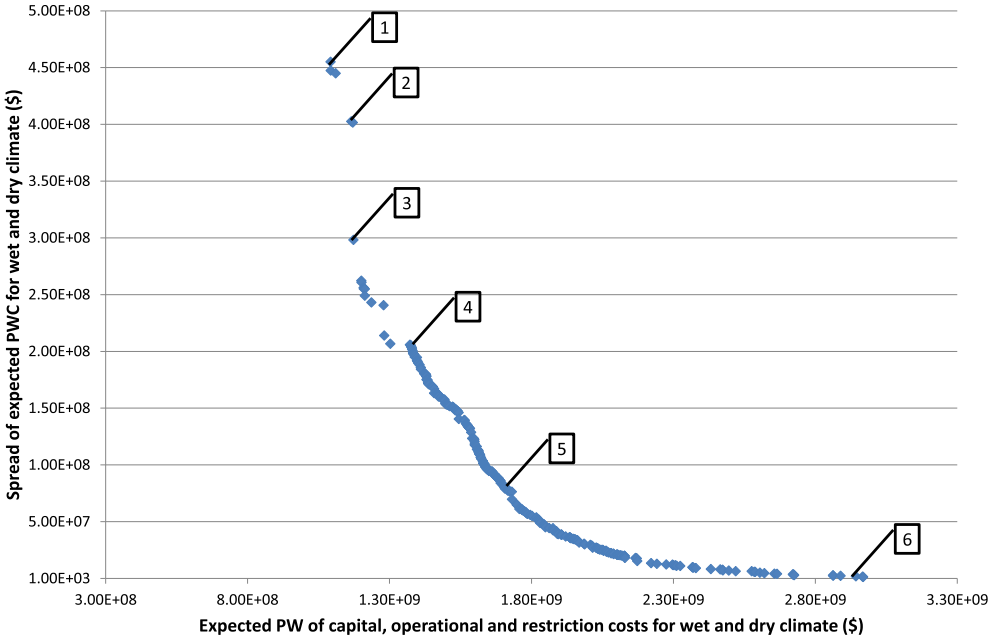


Fig. 5. Robust Pareto optimal front for the 200% × current demand scenario.

efficient, more robust) opt for the larger and more costly option.

c) The increasing investment in desalination is accompanied by a decreasing restriction trigger resulting in a lower frequency of restrictions and emergency rationing.

d) Solution 1 exhibits the greatest difference in restriction frequency between the dry and wet climates. Only solutions 5 and 6 (less efficient, more robust) produce restriction frequencies less than 5% for both the dry and wet climate scenarios, which is a common level of service criterion for Australian water utilities.

e) To reduce the cost sensitivity to a minimum requires an over-engineered and very conservatively operated system. Solution 6 opts for a large desalination plant with 125 ML/d capacity operated continuously to supply 32% of average daily demand. While this does virtually eliminate sensitivity to the dry and wet climate scenarios, the cost to achieve this level of risk aversion is patently excessive.

4.4. System sensitivity to uncertainty in 2070 future climate change

The construction of the future climate change scenarios described in Section 3.1 represented a state-of-the art application at the water resource management agency level. The selection of the better performing GCM models and the adoption of the 50th percentile estimate of local change in rainfall and PET per degree of global warming, were motivated by the need to define plausible limits on future climate change. These decisions relied on the exercise of judgement and therefore there remains a risk that future climate change could be more severe than tested in this study. Not only is there deep uncertainty about the scenario probabilities,

Table 9
Summary of labelled solutions on Pareto front in Fig. 5: 200% × current demand scenario.

	Solution	1	2	3	4	5	6
Objectives	Total expected PW of capital, operational and restriction costs ($m)	1092	1165	1173	1372	1968	2943
	Spread of total expected PW cost ($m)	455	402	298	205	31	1
Frequency	Chance of restrictions in any month (%) for dry and wet scenario	16.5 (dry) 6.3 (wet)	14.9 (dry) 5.6 (wet)	10.4 (dry) 3.4 (wet)	6.5 (dry) 2.0 (wet)	2.4 (dry) 0.82 (wet)	0.31 (dry) 0.13 (wet)
	Chance of emergency rationing in any month (%) for dry and wet scenario	1.0 (dry) 0.32 (wet)	0.91 (dry) 0.28 (wet)	0.71 (dry) 0.21 (wet)	0.32 (dry) 0.11 (wet)	0.055 (dry) 0.032 (wet)	0.0 (dry) 0.0 (wet)
Decisions	Level 1 restriction storage trigger (% of total storage)	39.4	39.4	33.1	28.1	25.3	19.9
	Restriction trigger increment (% of total storage)	5.7	5.4	5.1	5.1	5.1	5.1
	Emergency restriction triggers (% of total storage)	0.759 (12.6%)	0.751 (13.4%)	0.768 (9.7%)	0.681 (5.2%)	0.511 (2.5%)	0.641 (0%)
	Severe rationing demand scaling factor	0.678	0.677	0.696	0.686	0.697	0.546
	Storage trigger to initiate desalination plant construction	–	–	–	–	–	–
	Desalination plant capacity (ML/day)	17	17	17	50	74	125
	Desalination plant construction lead time (month)	0	0	0	0	0	0
	Desalination plant trigger (% of total storage)	49.7	49.3	49.3	59.1	100	99.7
	New reservoir capacity (ML)	229,918	229,953	229,934	229,929	229,918	222,930
	Chichester to Newcastle pipeline capacity expansion (ML/day)	8	12	6	2	7	8
	Grahamstown base gain	8623	8101	8590	8719	8117	9623
	Grahamstown incremental gain	32	59	47	56	69	34
	Grahamstown pumping trigger (% of total storage)	98.9	99.7	99.8	99.2	80.8	73.5
	Number of domestic rainwater tanks	36	25,050	50,232	91	245	50,223

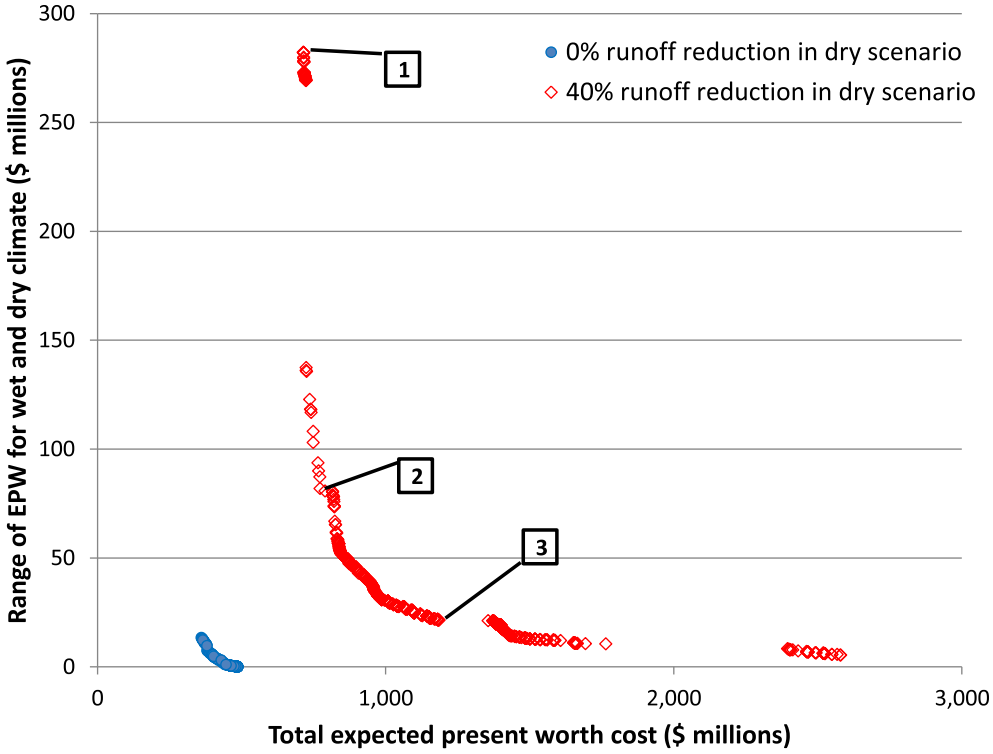


Fig. 6. Comparison of Pareto front for 128% × current demand scenario for dry climate scenario runoff reduced by 0% and 40%.

there is also additional uncertainty about the ability of the dry and wet scenarios to define plausible limits.

To reflect on the significance of this uncertainty, the 128% × current demand scenario was rerun with the runoff in the dry scenario reduced by 40%. Fig. 6 compares the Pareto fronts for 0% and 40% reduction in runoff for the dry climate change scenario, while Table 10 summarizes the three solutions labelled on the 40% runoff reduction Pareto front. For the 40% reduction case, there is a steep trade-off between solutions 1 and 2 – for a $58m increase in total EPW cost, there is a $200m reduction in cost spread. All three labelled solutions opt for the maximum reservoir capacity and permanent desalination, albeit using the small desalination plant. The main differences between the three solutions occur in the choice of restriction, pumping and desalination triggers and the number of installed rainwater tanks. Of these four decisions, three are operational which offers the water agency considerable flexibility.

It was argued in Section 2.2 that in the presence of deep uncertainty the trade-off between measures of efficiency and robustness involves subject reflection about the uncertainty in the scenario probabilities. This analysis suggests that the decision maker may need to also reflect on the uncertainty about the worst-case scenarios actually representing the worst case. While the sensitivity analysis does not resolve how severe the dry future climate change scenario may be, reflection on the results of the sensitivity analysis may help to identify the characteristics of solutions that can cope with the occurrence of such an unfavourable scenario. In this case study, the characteristics associated with solutions that cope include maximized reservoir capacity and adoption of permanent but small-scale desalination with the fine tuning

Table 10
Summary of labelled solutions on Fig. 6: 40% reduction in dry state runoff, 128% × current demand scenario.

	Solution	1	2	3
Objectives	Total expected PW of capital, operational and restriction costs for wet and dry climate ($m)	715	773	1182
	Range of expected PW cost for wet and dry climate ($m)	282	82	22
Decisions	Level 1 restriction storage trigger (% of total storage)	45	25	23
	Restriction trigger increment (% of total storage)	10	5	5
	Emergency restriction trigger	0.70	0.66	0.69
	Severe rationing demand scaling factor	0.53	0.65	0.65
	Storage trigger to initiate desalination plant construction	–	–	–
	Desalination plant capacity (ML/day)	14.8	19.55	19.6
	Desalination plant construction lead time (month)	0	0	0
	Desalination plant trigger (% of total storage)	61	94	94
	New reservoir capacity (ML)	229,987	229,949	229,893
	Chichester to Newcastle pipeline capacity expansion (ML/day)	0	0	0
	Grahamstown base gain	8756	8890	8405
	Grahamstown incremental gain	42	21	21
	Grahamstown pumping trigger (% of total storage)	100	50	50
	Number of domestic rainwater tanks	258	24	93,791

left to tweaking operational decisions and domestic rainwater tank installation, both of which offer considerable flexibility in their execution.

4.5. Using an asymmetric robustness measure

In Section 2.3 it was noted that the use of spread between worst and best cases as a robustness measure "equally" penalizes deviations above or below the expected value. In contrast, making the worst case cost as the robustness criterion only penalizes "undesirable" deviations. Because the two climate states (dry and wet) have been assigned equal probability, it can be readily shown that

$$EPW_{dry} = EPW + \frac{Sp}{2} \quad (6)$$

where EPW_{dry} is the expected present worth cost for the dry climate state, Sp is the spread or difference in costs between dry and wet climate states, and EPW is the present worth cost averaged over the dry and wet states. Table 11 summarizes EPW, Sp and EPW_{dry} for the results presented in the previous sections.

If the decision maker is interested in minimizing the worst case (or dry state) EPW cost, it is clear that the solution that minimizes the EPW cost over both scenarios also minimizes the dry state EPW cost for both the 128% and 200% × current demand scenarios. There is no trade-off. However, for the 128% × current demand scenario with a 40% runoff reduction in the dry state, there is a trade-off between solutions 1 and 2 – the dry state cost can be reduced from $856m to $814m in return for an increase in EPW from $715m to $773m.

In this case study the cost of reducing sensitivity (as measured by the spread) to low levels is very high and requires solutions that; it would be difficult to justify the high degree of over-engineering required to achieve a low cost spread. This behaviour is controlled by the available options and the costs of "normal" and emergency restrictions. Only in the 128% × current demand scenario with dry state runoff reduced by 40%, do we see a steep trade-off between spread and EPW cost (between solutions 1 and 2 in Fig. 6). The inclusion of the asymmetric robustness measure draws attention to this issue, a point strongly made by Ray et al. (2013). Mindful of the fact that the results are case-specific, it would nonetheless seem prudent to explore contrasting robustness measures.

5. Conclusions

Urban bulk water systems are typically required to supply water with high reliability and, in the event of extreme drought, maintain sufficient supply to enable the urban area to avoid catastrophic economic and social collapse. In view of the deep uncertainty about future climate change and the vulnerability of bulk water systems to droughts with return periods in excess of the design return period, it is vital that robust solutions be found that secure urban bulk water systems against extreme drought.

This study has approached this challenge by integrating several approaches. To account for the drivers affecting extreme drought, a stochastic model of multi-site streamflow conditioned on future climate change scenarios was developed. To enable the decision maker to trade-off efficiency against sensitivity to climate change scenarios, the Monte Carlo simulation approach of Mortazavi et al. (2012) was incorporated into the Mulvey et al. (1995) robust optimization framework. To maximize trade-off opportunities, a comprehensive decision space was adopted to include operating rules, investment in new sources and source substitution and a drought contingency plan with multiple actions with increasingly severe economic and social impact. Our approach attempts to give expression to the viewpoint presented by Weaver et al. (2013, p.39) who argue that climate models should not viewed as "prediction machines within 'predict-then-act' decision frameworks, but as scenario generators, sources of insight into complex system behaviour, and aids to critical thinking within robust decision frameworks".

The Lower Hunter case study was used to demonstrate the capability of the robust optimization to assist water managers reflect on the consequences of uncertain future climate change and select solutions that adequately cope. The sensitivity of solutions, as measured by the spread or difference between dry and wet climate scenario costs, to uncertainty about future climate change varied considerably. For the 128% × current demand scenario, with a new reservoir included as an option, there was little sensitivity to the dry and wet climate scenarios resulting in a Pareto solution set spanning a narrow range of trade-offs; however, when the new reservoir was excluded from the list of available options, a significantly inferior Pareto front resulted with considerable trade-offs between total expected present worth cost and spread. For the 200% × current demand scenario, the high demand load resulted in a very considerable trade-off between total expected present worth cost and spread.

Table 11
Comparison of the spread and dry state EPW cost robustness measures.

Scenario	Table	Solution	EPW $m	Spread S $m	Dry state EPW_{dry} $m
128% × current demand (reservoir option excluded)	7	1	812	102	863
		2	972	61	1003
		3	1225	40	1245
		4	1833	20	1843
		5	3399	7	3403
128% × current demand (reservoir option included)	8	6	360	13	367
		7	392	6	395
		8	487	0	487
200% × current demand	9	1	1092	455	1320
		2	1165	402	1366
		3	1173	298	1322
		4	1372	205	1475
		5	1968	31	1984
		6	2943	1	2944
128% × current demand (40% runoff reduction in dry state)	10	1	715	282	856
		2	773	82	814
		3	1182	22	1193

This variability in outcomes highlights the fact that (possibly large) uncertainty about future climate change may not necessarily produce significantly different performance trajectories. The key consideration is that of the exogenous stress imposed on the system. This not only reflects the obvious stressors, namely population (and consequent demand) and future climate change uncertainty, but also the size and scope of the decision space which determines the possibilities explored by the optimizer. For example, if the new reservoir option were not available to add more storage capacity to the system or if the desalination option were not available to provide a climate-independent source of water, the stress level would be considerably higher probably producing very different performance outcomes and solution sets. Indeed many Pareto optimal solutions did hit the upper limit on the reservoir capacity suggesting that making available more opportunities for storage could produce better outcomes.

When the worst-case cost (corresponding to the dry climate scenario) was selected as the robustness measure instead of cost spread, it was found that there was only one Pareto optimal solution, the least expected cost solution was also the minimum worst-case cost solution; moreover, as the spread was reduced, both the expected and worst-case costs increased. This was because the only way to reduce the cost spread between the dry and wet scenarios was to invest in large desalination capacity and operate it conservatively. This suggests there is merit in comparing robustness measures that penalize deviations in different ways, a conclusion that echoes the findings of Ray et al. (2013).

The complexity of the issues encountered in the case study highlight the fact that the robust optimization is at best a tool to help decision makers reflect in a structured manner on deep uncertainty about future climate change and its consequences. This reflection would no doubt consider the decision maker's metadata not explicitly captured in the formal objectives and constraints.

Finally it needs to be stressed that the robust optimization results are conditional on the credibility of the range of potential climate change impacts on rainfall and potential evapotranspiration PET and also on the credibility of stochastic modelling of natural climate variability. A sensitivity analysis on the severity of the dry climate scenario illustrated the importance of this conditioning. While the climate change factor approach used in this study was regarded as best practice at the time of writing (and the most feasible approach given the project's time, budget and GCM data availability constraints), alternative methods are available (as are updated emission scenario and GCM information associated with CMIP5 and the Intergovernmental Panel on Climate Change's fifth assessment report). These alternative emission scenarios, climate model outputs and approaches for developing climate change scenarios may yield different ranges and resolving this issue is considered an important goal in the quest for identifying robust climate change adaptation strategies in water resource planning.

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